

Matthew Jordan

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EDUCATION

The University of Alabama - Tuscaloosa, AL Aug 2024 - Present

Bachelor of Science Mechanical Engineering GPA 3.5 / 4.0

Bachelor of Science Aerospace Engineering GPA 3.5 / 4.0

- Dean's List (Spring 2025); Alpha Lambda Delta and Lambda Sigma Honor Societies.

ENGINEERING EXPERIENCE

Systems Engineer Lead - Crimson Racing (Formula SAE) 2026 - Present

- Lead systems-level integration across powertrain, aerodynamics, suspension, chassis, and vehicle dynamics subsystems for the team's 2027 Michigan competition car, coordinating trade studies and interface requirements between groups for both the IC and EV teams.
- Vehicle performance modeling in MATLAB, building lap-time and parameter sensitivity tools that quantify how mass, aerodynamic, suspension and powertrain changes drive competition scoring.
- Validating projected vehicle performance goals to the designed and manufactured vehicles.
- Providing overall system integration to balance performance with design tradeoffs that limit multiple sub teams

Cooling Systems Engineer - Crimson Racing (Formula SAE) 2025 - Present

- Design and analyze the thermal management (cooling) system, sizing components to manage engine heat rejection under sustained endurance-event loads.
- Supports manifold and powertrain design and fabrication, translating analysis into manufacturable hardware.
- Developed fluid property testing simulation to determine and prove the best fluid to use within FSAE regulations for an EV platform.

Team Member & Vehicle Simulation - Crimson Racing (Formula SAE) 2024 – Present

- Developed a calibrated vehicle-speed and performance estimation model that improved lap-simulation accuracy, validated against measured on-track (MoTeC) data.
- Built a fuel-consumption model integrating measured fuel-flow data, engine efficiency, manifold pressure, and engine rpm to predict efficiency-event scoring for Michigan IC through determining fuel consumption rate, validated against official competition efficiency results.
- Built a dynamic power limit model to test different power consumption cases to predict efficiency-event scoring for Michigan EV
- Manufactured EV cooling system through composite and additive manufacturing techniques
- Designed an automated documentation system that logs files, creates short papers, and categorizes file locations to accelerate onboarding and finding important documents

ADDITIONAL EXPERIENCE & LEADERSHIP

FAA-Certified UAV Pilot / Aerial Photographer (freelance) 2023 – Present

- Licensed Remote Pilot in Command (FAA Part 107); plan and fly UAV missions and deliver high-resolution aerial imagery for clients, balancing safety, airspace, and technical constraints.

Treasurer - Engineering Leadership & Development Society 2025 – 2026

- Managed organization budget and finances while supporting professional-development programming for engineering students.
- Planned funds for inclusive and diverse workshops that taught students engineering skills
- Guided students through Microsoft office to develop their management skills

TECHNICAL SKILLS

Programming & Analysis: MATLAB, Python, Fortran; numerical methods, simulation, data analysis.

Design & Manufacturing: SolidWorks; Siemens NX; Fusion 360; GT-SUITE; manual & CNC lathe machining; GD&T-driven part design.

Domain: Vehicle dynamics & lap simulation, powertrain, thermal/cooling systems, aerodynamics trade studies, computational physics.

Tools & Certifications: MoTeC data analysis; Microsoft Office Specialist - Excel Associate; FAA Part 107 Remote Pilot Certificate.